

Ray Pomponio

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PubMed publications

EDUCATION

MS, Biostatistics May 2023
University of Colorado–Denver *Thesis: Testing for mean differences in partially matched data*

BS, Economics May 2018
University of Pennsylvania, Wharton School

TECHNICAL SKILLS

Programming: Python (NumPy, Scikit-learn, SciPy, PyTorch, Matplotlib), R (tidyverse, mice, ggplot2), SQL, SAS, Unix shell scripting

ML & AI: Deep learning (CNNs, brain age modeling), supervised/unsupervised ML, LLM-assisted workflows (Claude, Copilot), image processing, computer vision, Monte Carlo simulation

Statistics: Multivariate regression, longitudinal modeling, Bayesian data analysis, causal inference, survival analysis, missing data/multiple imputation, sample size/power analysis

Data Infrastructure: SQL, AWS, HPC environments (parallel job scheduling), Docker, DBT, REDCap (API automation), Power BI, Git, end-to-end data pipeline design

Neuroimaging: FreeSurfer, sMRIPrep, MRIQC, neuroHarmonize (author), ComBat-based harmonization

RESEARCH & PROFESSIONAL EXPERIENCE

Data Scientist / Lead Biostatistician 2024 – Present
University of Pittsburgh / Children’s Hospital of Pittsburgh

- Lead statistician on a **CDC-sponsored New Vaccine Surveillance Network** study; designed and executed the full clinical analytics pipeline across 8,000+ pediatric encounters, from study design through publication.
- Built end-to-end data infrastructure integrating multi-system clinical sources including CRFs in REDCap, EHR extracts, and laboratory read-outs; developed Power BI dashboard (ranked 834 of 3,571 UPMC reports) and automated recurring query workflows, saving coordinators 3–4 hours/week.
- Developed and validated **multivariate risk-stratification models** (logistic regression, ordinal outcomes) to predict severe respiratory illness in pediatric populations; results informed CDC public health guidance.
- Integrated LLM-assisted workflows (Claude, Copilot) into report drafting, slide narratives, and code generation; mentored junior faculty and fellows, contributing to 3 successful grant proposals and one NIH (K) award won.

Independent Statistical Consultant 2023 – 2024
Upwork.com

- Advised medical device clients on **non-inferiority endpoints**; performed sample size and power calculations, drafted statistical analysis plans (SAPs) and contributed to **regulatory submissions** at FDA.

Graduate Research Assistant 2021 – 2023
University of Colorado–Denver, Department of Biostatistics & Informatics

- Collaborated with multidisciplinary therapeutic teams to analyze outcomes in pulmonary and critical care medicine; analyzed **clinical trial data** in asthma and allergy treatment using IL-5 biologics.
- Evaluated exploratory biomarker data in **lung transplant recipients**; demonstrated predictive power of cell-free DNA measures in hosts with clinical endpoints including allograft instability and transplant rejection.

Data Analyst 2018 – 2020
University of Pennsylvania, School of Medicine — Davatzikos Lab (CBICA)

- Designed and deployed **neuroHarmonize**, a production-stage open-source Python package (100+ GitHub stars, **500+ citations**) for harmonizing multi-site neuroimaging datasets using ComBat-GAM statistical methods; adopted as a standard preprocessing tool by neuroimaging consortia.
- Built automated **ML pipelines for MRI image processing and feature extraction**, reducing preprocessing timelines from months to days for a multi-center neurodegeneration biomarker consortium.
- Applied **deep learning and computer vision** (CNNs) to 14,000+ MRI scans to model brain age trajectories and disentangle neurodegeneration from typical aging; results published in *Brain* and *Alzheimer’s & Dementia*.

AWARDS & HONORS

- Certificate of Appreciation, Centers for Disease Control and Prevention, 2024–2026.
- Marvin Porter Award for Outstanding Academic Achievement, UC Denver Biostatistics, 2023.
- Dean’s List, University of Pennsylvania Wharton School, 2018.
- Hackathon Winner, Ryder Transportation analytics, Wharton Customer Analytics Initiative, 2018.

SELECTED PUBLICATIONS

Full list: ncbi.nlm.nih.gov/myncbi/raymond.pomponio.1/bibliography/public/

Neuroimaging & Harmonization

- **Pomponio R**, Erus G, Habes M, et al. Harmonization of large MRI datasets for the analysis of brain imaging patterns throughout the lifespan. *NeuroImage*. 2020;208:116450.
- Bashyam VM, Erus G, Doshi J, . . . , **Pomponio R**, et al. MRI signatures of brain age and disease over the lifespan based on a deep brain network and 14,468 individuals worldwide. *Brain*. 2020;143(7):2312–2324.
- Habes M, **Pomponio R**, Shou H, et al. The Brain Chart of Aging: machine-learning analytics reveals links between brain aging, white matter disease, amyloid burden, and cognition in the iSTAGING consortium of 10,216 harmonized MR scans. *Alzheimer’s & Dementia*. 2021;17(1):89–102.
- Nasrallah IM, Gaussoin SA, **Pomponio R**, et al. Association of intensive vs standard blood pressure control with MRI biomarkers of Alzheimer disease: secondary analysis of the SPRINT MIND randomized trial. *JAMA Neurology*. 2021;78(5):568–577.
- Chen AA, Srinivasan D, **Pomponio R**, et al. Harmonizing functional connectivity reduces scanner effects in community detection. *NeuroImage*. 2022;256:119198.

Clinical & Translational Research

- **Pomponio R**, Peterson RA, Owusu M, et al. Phosphatidylethanol measures in patients with severe COVID-19-associated respiratory failure identify a subset with alcohol misuse. *Alcoholism: Clinical and Experimental Research*. 2025;49(1):165–174.
- Pless LL, Damodaran L, **Pomponio R**, et al. Emergence and epidemiology of dominant variants of human metapneumovirus in the United States between 2016 and 2021. *mBio*. 2026;17(2):e0261925.
- Dunbar PJ, Peterson R, McGrath M, **Pomponio R**, et al. Analgesia and sedation use during noninvasive ventilation for acute respiratory failure. *Critical Care Medicine*. 2024;52(7):1043–1053.
- Smith JB, Peterson RA, **Pomponio R**, et al. Donor derived cell free DNA in lung transplant recipients rises in setting of allograft instability. *Frontiers in Transplantation*. 2024;3:1497374.

Peer reviewer: *Imaging Neuroscience, NeuroImage* (5+ manuscripts). Grant support: CDC New Vaccine Surveillance Network, 2024–present (PI: Marian Michaels, MD).